

Temperature compensation in FBG sensing: a survey

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Abstract. Fibre Bragg gratings answer to strain and to temperature alike, and every sensing system has to tell the two apart. This survey collects the three families of compensation in use: a reference grating kept free of strain, a packaged athermal grating, and the dual-wavelength method that fits a two-by-two matrix once and inverts it thereafter. Each is measured against the same question: how far the recovered strain drifts over a day of weather.

The dual-wavelength method is the cheapest to install and the most often misread. Outside the range over which the coefficients were fitted, the inverse matrix is an extrapolation: the photo-elastic and thermo-optic coefficients are not linear there, and the recovered strain drifts with temperature. Within the fitted range the method holds to a few microstrain, which is as good as the packaged gratings and an order of magnitude cheaper.

Fast transients are the second limit. A cold front moving across a span leaves two gratings a few metres apart at different temperatures for some minutes, and every method surveyed assumes they are not.